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F. Digit Lock

time limit per test: 1 second

memory limit per test: 256 megabytes

Alice has a treasure secured by a 4-digit numeric lock.



The 4-digit numeric lock.

Of course, Alice knows the correct combination C required to unlock this lock. However, there is an issue. Due to a mechanical fault in the numeric lock, certain combinations are deemed **forbidden**.

Alice wants to unlock this lock. In a single step, Alice can take one of the four "wheels" and rotate it by one step. More formally, Alice can take a single digit and either increment it or decrement it by 1 (modulo 10). The digits rotate around like clocks, so if we decrement 0 the digit becomes 9 and if we increment 9 the digit becomes 0. See the sample explanation for better understanding.

Alice wants to find the minimum number of steps required to reach the correct combination C from the starting combination S .

Input

The first line of the input contain the starting combination S and the correct combination C . Both are described in the input as 4-digit numbers. It is guaranteed that the starting combination is not forbidden.

The second line contains an integer n ($1 \leq n \leq 10^4$), the number of forbidden combinations.

The next n lines each contain a forbidden combination.

Output

If it is impossible to reach C , then output -1 . Otherwise, output the minimum number of required steps.

Examples

input

Copy

0000 0003

CSE221 Assignment 06 Summer 2026

Contest is running

3 days

Contestant



→ Submit?

 Language: Java 21 64bit

 Choose file: Choose File No file chosen


Success!



Submit

1 0001	
output	Copy
5	

input	Copy
0000 0007 1 0006	
output	Copy
3	

input	Copy
1234 5678 12 2234 3234 4234 5234 1334 1434 1534 1634 1244 1254 1264 1274	
output	Copy
16	

Note

In the first sample, our starting combination is 0000 and our target combination is 0003, while there is only one forbidden combination: 0001. So we can not directly increment the last digit in our starting combination. The following steps are optimal:

0000 → 0010 → 0011 → 0012 → 0013 → 0003.

in the second sample, our starting combination is again 0000 while our target combination is 0007. The following steps are optimal: 0000 → 0009 → 0008 → 0007.

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